QPCODE: .....

Reg. No.: .....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION MONTH 2026

FIFTH SEMESTER

DSE-MG5DSEBIF301 – SOFT COMPUTING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks:** 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each Course Outcome to enhance their overall outcome attainability.

Part A – Short Answer Questions

Answer any 5 questions. Each question carries 2 marks.

1. What is Soft Computing? Mention any two areas in which it is applied. [U][1]

2. State any four features of Artificial Neural Networks (ANNs). [U][1]

3. What is a fuzzy set? How does it differ from a classical set? [U][3]

4. Define membership function in fuzzy logic. Mention any two types. [U][3]

5. What is an Evolutionary Algorithm? [U][4]

6. Define dimensionality reduction. Why is it useful in Bioinformatics? [U][2]

7. What is feature selection in pattern recognition? [U][5]

[5 × 2 = 10]

Part B – Short Essay Questions

Answer any 6 questions. Each question carries 5 marks.

8. Explain the history, biological motivation and major areas of Soft Computing. [U][1]

9. Explain supervised and unsupervised neural networks with suitable examples. [U][1]

10. Explain the structure and working of a Perceptron. How is a Multi-Layer Perceptron different? [A][1]

11. Explain the properties of fuzzy sets and the basic fuzzy operations. [U][3]

12. Explain membership functions and their role in fuzzy logic with suitable examples. [U][3]

13. Describe the major components of an Evolutionary Algorithm. [U][4]

14. Explain Genetic Algorithms and their basic steps. [A][6]

15. Explain clustering, classification and dimensionality reduction as pattern recognition techniques in Bioinformatics. [A][5]

[6 × 5 = 30]

Part C – Long Essay Questions

Answer any 3 questions. Each question carries 10 marks.

16. Explain Artificial Neural Networks in detail. Discuss their features, supervised and unsupervised learning, Perceptron, Multi-Layer Perceptron and the relevance of ANNs in Bioinformatics. [E][1]

17. Explain Fuzzy Logic in detail. Discuss fuzzy sets, properties, membership functions, fuzzy operations and applications, with suitable examples. [E][3]

18. Explain Evolutionary Computing in detail. Discuss Evolutionary Algorithms, their components, Genetic Algorithms, Evolutionary Programming and Genetic Programming. [E][4]

19. Explain Pattern Recognition in Bioinformatics. Discuss clustering, dimensionality reduction, classification, feature selection and their applications. [A][5]

20. Discuss the role of Soft Computing techniques in Bioinformatics. Explain how ANN, Fuzzy Logic and Evolutionary Computing can be applied to solve Bioinformatics problems. [A][1]

[3 × 10 = 30]